UNIVERSITY OF LONDON

BSc EXAMINATION 2025

For Internal Students of
Royal Holloway

DO NOT TURN OVER UNTIL TOLD TO BEGIN

IY3660: Applications of Cryptography

Time Allowed: **TWO hours**

Attempt **THREE** questions
Please answer each question on a separate page
No credit will be given for attempting any further questions

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Question 1 - Provable Security

In the context of cryptography, security definitions are fundamental concepts.

(a) Define Indistinguishability under Chosen Plaintext Attack (IND-CPA) for asymmetric encryption. Explain the steps involved in the IND-CPA game. [8 marks]

(b) Consider an encryption scheme where the ciphertext C is generated as $C = E_{pk}(M)||M$, where $E_{pk}(M)$ is an IND-CPA secure encryption of message M under public key pk, and $||$ denotes concatenation. Is this scheme IND-CPA secure? Justify your answer. [5 marks]

(c) Compare IND-CPA security with IND-CCA (Indistinguishability under Chosen Ciphertext Attack) security. Name a scheme which is IND-CPA secure but not IND-CCA secure. [3 marks]

(d) Define the security property of Unforgeability under Chosen Message Attack (UF-CMA) for a digital signature scheme (KGen, Sign, Vrf). Explain the steps involved in the UF-CMA game. [8 marks]

(e) Consider a signature scheme where the signature σ is generated as $\sigma = \mathsf{Sign}_{sk}(M)||M$, where $\mathsf{Sign}_{sk}(M)$ is the signature algorithm of a UF-CMA secure signature scheme, sk is the signing key and $||$ denotes concatenation. Is this scheme UF-CMA secure? Justify your answer. [5 marks]

(f) Name an example of a UF-CMA secure signature scheme, and explain what security service this notion intuitively provides. [5 marks]

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Question 2 - Symmetric cryptography

Answer the following questions related to symmetric cryptography.

(a) Define what a block cipher is and explain how it differs from a stream cipher. [6 marks]

(b) Consider a simple block cipher with block size of 4 bits. Let $K = 1010$ be the secret key.

i. Let the encryption process consist of XORing the plaintext block with the key. Encrypt the plaintext blocks $[1100, 0011]$ using the cipher. [4 marks]

ii. Encrypt the same plaintext blocks using CBC mode with Initialization Vector IV$= 0110$. [8 marks]

In both cases, show your calculations step by step.

(c) Define what a Message Authentication Code (MAC) is, and name two common approaches for constructing MAC algorithms, briefly explaining how they work. [11 marks]

(d) Explain what Authenticated Encryption (AE) is and state two key security notions for AE. [5 marks]

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Question 3 - Public-key cryptography

Answer the following questions related to public-key cryptography.

(a) Explain the purpose of the Diffie-Hellman Key Exchange protocol and describe the protocol in detail, specifying the setup, the parties involved, the exchanged messages and the final output. You may use a diagram to specify the protocol. [9 marks]

(b) Discuss the security of the Diffie-Hellman protocol. In particular, identify and describe what is the security assumption it relies on, and discuss whether the protocol in its basic form is vulnerable to any attacks. [8 marks]

(c) Choose a version of RSA-based signature scheme. Describe it in detail and discuss its security. [8 marks]

(d) Describe what Elliptic Curve Cryptography (ECC) is, and how it relates to public-key cryptography. Explain why ECC is considered more efficient than RSA for the same security level. [9 marks]

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Question 4 - Applications of cryptography

Transport Layer Security (TLS) is a widely used cryptographic protocol for securing communication over the internet. Write an essay discussing the following aspects of TLS:

- The main goals of TLS and why they are important in securing communication.
- The cryptographic components of TLS, explaining the role of symmetric and asymmetric cryptography.
- The handshake process in TLS, focusing on how it establishes a secure session.

Justify your discussion to demonstrate your understanding of the protocol.

[34 marks]

END

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